

ASSIGNMENT 3.3

Q1. Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page .Hint: If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.

Steps to Open the Wi-Fi Router Login Page

Step 1: Connect to Wi-Fi

Connect your computer or laptop to your home Wi-Fi network.

Step 2: Open Command Prompt

Press Windows + R, type cmd, and press Enter.

Step 3: Find the Default Gateway

In Command Prompt, type:

Ipconfig

Press Enter and find Default Gateway under your Wi-Fi adapter.

Example:

Default Gateway. . . : 192.168.1.1

Step 4: Open the Router Login Page

Open Chrome, Edge, or another web browser. Enter the Default Gateway address in the address bar.

For example:

http://192.168.1.1

If your Default Gateway is 192.168.0.1, use:

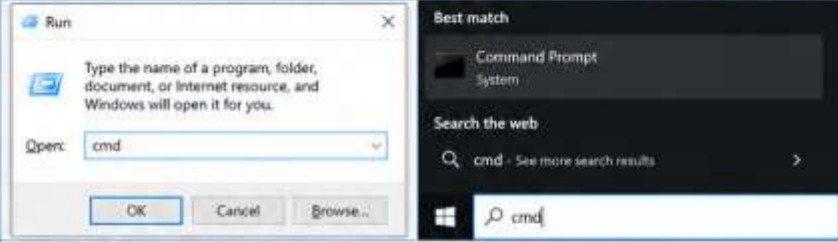
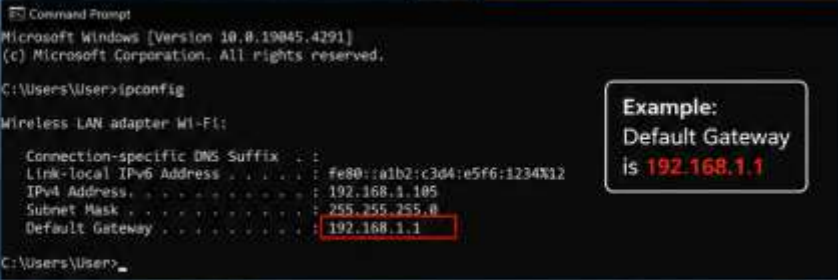
http://192.168.0.1

Step 5: View the Login Page

The router's administrator login page should appear. It may ask for a username and password.

Step 6: Take a Screenshot

Take a screenshot showing the router login page. Do not include or share your router password.

1 Connect to Wi-Fi Connect your computer or laptop to your home Wi-Fi network.	
2 Open Command Prompt Press Windows + R, type cmd, and press Enter.	
3 Find the Default Gateway In Command Prompt, type ipconfig and press Enter. Find Default Gateway under your Wi-Fi adapter.	
4 Open the Router Login Page Open a web browser and enter the Default Gateway address in the address bar. Example: http://192.168.1.1	
5 View the Login Page The router's administrator login page will appear. It may ask for a username and password.	

Q2. Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

Step 1: Connect your laptop to your Wi-Fi.

Step 2: Press **Windows + R**.

Step 3: Type:

cmd

and press **Enter**.

Step 4: Type:

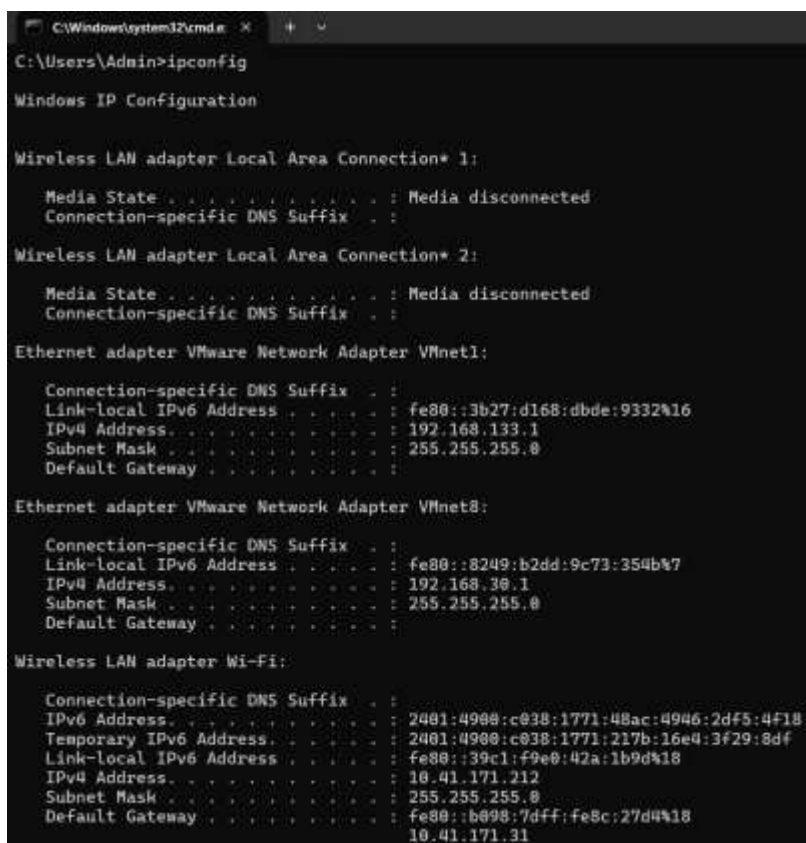
ipconfig

and press **Enter**.

Step 5: Find the **Wireless LAN adapter Wi-Fi** section.

You will see something like:

IPv4 Address.....: 10.41.171.212
Subnet Mask.....: 255.255.255.0
Default Gateway.....: 10.41.171.31



```
C:\Windows\system32\cmd.exe
C:\Users\Admin>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . :
    Link-Local IPv6 Address . . . . . : fe80::3b27:d168:dbde:9332%16
    IPv4 Address. . . . . : 192.168.133.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix  . :
    Link-Local IPv6 Address . . . . . : fe80::8249:b2dd:9c73:354b%7
    IPv4 Address. . . . . : 192.168.30.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2401:4908:c038:1771:48ac:4946:2df5:4f18
    Temporary IPv6 Address. . . . . : 2401:4908:c038:1771:217b:16e4:3f29:8df
    Link-Local IPv6 Address . . . . . : fe80::39c1:f9e0:42a:1b9d%18
    IPv4 Address. . . . . : 10.41.171.212
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::b098:7dff:fe8c:27d4%18
                                10.41.171.31
```

Q3. Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access. Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.

Edit network IP settings

☒ On

IP address
10.41.171.250

Subnet mask
255.255.255.0

Gateway
10.41.171.31

Preferred DNS
8.8.8.8

DNS over HTTPS
Off

Alternate DNS
1.1.1.1

DNS over HTTPS
Off

Save Cancel



Connect to the internet

You're offline. Check your connection.

Retry

IP assignment:	Automatic (DHCP)
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DNS server assignment:	Automatic (DHCP)
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Step 1: Open Network Adapter Settings

1. Press Windows Key + R, type `ncpa.cpl`, and press Enter.
2. Right-click your active Wi-Fi network connection and select Properties.
3. Double-click Internet Protocol Version 4 (TCP/IPv4).

Step 2: Change to Static IP

1. Select Use the following IP address.
2. Enter the following details based on your router's gateway:
 - IP Address: 10.41.171.250
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 10.41.171.31
3. Enter 8.8.8.8 for the Preferred DNS server.
4. Click OK to save changes.

Step 3: Test Internet Connectivity

1. Open Command Prompt (`cmd`).
2. Run the command: `ping 8.8.8.8` or `ping 10.41.171.31`.
3. Observation: Internet connection failed (Request timed out / Destination host unreachable) because the network enforces DHCP and blocks manual static IPs.

Step 4: Revert Back to DHCP (Constraint)

1. Reopen `ncpa.cpl` > Right-click Wi-Fi > Properties > TCP/IPv4.
2. Select Obtain an IP address automatically.
3. Select Obtain DNS server address automatically.
4. Click OK to restore internet connection.

Q4. In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

The screenshot shows the DHCP Settings page of a TP-Link TL-WR840N router. The DHCP Server is enabled. The IP Address Pool is set from 192.168.0.100 to 192.168.0.199. Other settings include Address Lease Time of 120 minutes, Default Gateway of 192.168.0.1, and Primary DNS of 8.8.8.8. A 'Save' button is at the bottom.

Field	Value	Notes
DHCP Server	Enable	
Start IP Address	192.168.0.100	
End IP Address	192.168.0.199	
Address Lease Time	120	minutes (1-2880 minutes, the default value is 120)
Default Gateway	192.168.0.1	(Optional)
Default Domain		(Optional)
Primary DNS	8.8.8.8	(Optional)
Secondary DNS	8.8.4.4	(Optional)

Current DHCP Range:

Start IP Address: 192.168.0.100

End IP Address: 192.168.0.199

(Total: 100 IP addresses available)

What happens if two devices are assigned the same IP address?

If two devices have the same IP address, they will conflict with each other on the network. This can cause connectivity issues such as inability to access the internet, network disconnections, or intermittent access, because the router gets confused about which device to communicate with.